

A CMOS Oversampling D/A Converter with a Current-Mode Semidigital Reconstruction Filter

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Abstract—Oversampling digital-to-analog (D/A) converters employing sigma-delta modulation noise shaping and single-bit quantization are attractive for use in digital audio applications because of their relaxed reconstruction filtering requirements and their tolerance of component mismatch. However, the use of a two-level D/A interface results in a large amount of out-of-band quantization noise that typically must be attenuated by a carefully designed analog reconstruction filter. This paper introduces a means of simplifying the reconstruction filter design through the use of a semidigital finite-impulse-response (FIR) filter. In particular, it describes an oversampling D/A converter wherein a current-mode semidigital reconstruction filter is used to implement a multilevel D/A interface that attenuates the out-of-band quantization noise without requiring precise component matching. An experimental implementation of the converter achieves a dynamic range of 94 dB and 72 dB attenuation of out-of-band quantization noise for a baseband of 20 kHz. The prototype converter, which consists of a linear interpolator, a second-order noise shaper, and a 128-tap semidigital FIR filter, dissipates 59 mW from a 5-V supply and occupies an active area of 3 mm^2 when integrated in a $1.2\text{-}\mu\text{m}$ digital CMOS technology.

I. INTRODUCTION

THE growing demand for digital audio technology in consumer and communication electronics has created a need for high-resolution analog-to-digital (A/D) and digital-to-analog (D/A) converters that can be integrated together with digital processors in a standard CMOS technology. Oversampling D/A converters employing sigma-delta ($\Sigma\Delta$) modulation provide a cost-effective means of satisfying this need by exchanging circuit speed for amplitude resolution. In such converters, robust digital circuits are used to increase the sampling rate and shape the quantization noise of a digitally encoded signal, thereby easing the filtering and component matching requirements in the subsequent analog circuits.

Shown in Fig. 1 is the architecture for a D/A converter in which oversampling is used to remove low-order spectral images of the Nyquist-rate signal and thereby relax the transition-band requirements in the analog reconstruction filter. Noise shaping based on digital $\Sigma\Delta$ modulation with single-bit quantization is used to reduce the actual D/A conversion to a two-level interface that generates an inherently linear analog output with a large amount of quantization noise outside the bandwidth of interest [1]. Although the quantization noise in such a converter is shaped in frequency so that most of its

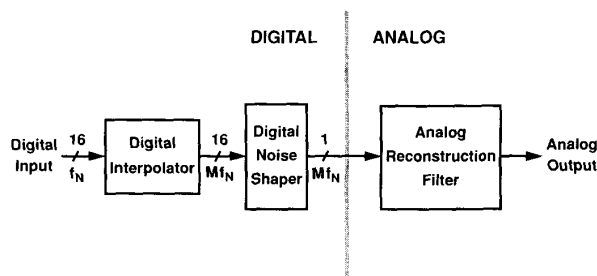


Fig. 1. Oversampling D/A converter with analog reconstruction filtering.

energy lies well above the audio frequency range, this noise must be attenuated by a carefully designed analog low-pass filter in order to avoid intermodulation and phase distortion.

Because the oversampling and noise shaping operations are both implemented with digital circuits, the performance of a conventional oversampling D/A converter is typically limited by circuit nonidealities in the analog reconstruction filter. This low-pass filter is generally implemented using discrete-time switched-capacitor and/or continuous-time RC circuits. Switched-capacitor low-pass filters are amenable to monolithic integration in a CMOS technology; however, because it is difficult to achieve the large dynamic range required in digital-audio applications, their use has generally been restricted to applications that demand a minimum number of external components [2]. Higher levels of performance can be obtained with continuous-time RC filters that employ external resistors, capacitors, and operational amplifiers [3], [4].

The design of the analog reconstruction filter in an oversampling D/A converter can be simplified if the out-of-band quantization noise is reduced by employing a multibit D/A interface together with a multibit quantizer in the digital noise shaper. With such an approach, the reconstruction filter can be reduced to a single-pole RC filter [5]. However, the multibit D/A interface circuit must meet the linearity requirements of the overall converter.

This paper describes a semidigital reconstruction filter for use with single-bit noise shaping that provides reduced quantization noise comparable to that obtainable with a multibit interface without requiring precise component matching. Current-switching circuits have been used to enable implementation of the filter in a standard digital CMOS technology, without the need for on-chip capacitors. An experimental realization of a linear interpolator, a 1-b noise shaper and a semidigital reconstruction filter has been integrated in a $1.2\text{-}\mu\text{m}$ CMOS technology. This circuit achieves a dynamic range

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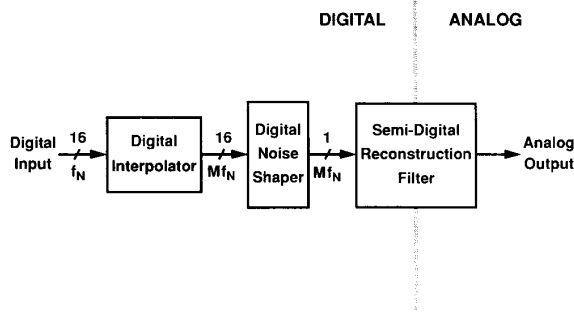


Fig. 2. Oversampling D/A converter with a semidigital reconstruction filter.

of 94 dB for a signal bandwidth of 20 kHz while attenuating the out-of-band noise energy by 72 dB.

Section II outlines the overall design of an oversampling D/A converter that employs semidigital reconstruction filtering. The architecture and design considerations for the semidigital filter are then described in Section III. Details of the CMOS circuit implementation of an audio-band oversampling D/A converter are presented in Section IV, and the experimental results are summarized in Section V.

II. OVERSAMPLING D/A CONVERTER

The architecture of an oversampling D/A converter with semidigital reconstruction filtering is shown in Fig. 2. This architecture consists of three major blocks: a digital interpolator, a digital noise shaper, and a semidigital reconstruction filter. Idealized representations of the spectra of signals at various stages in the converter are shown in Fig. 3. A 16-b digital word at the Nyquist sampling rate, f_N , of 44.1 kHz is upsampled by the oversampling ratio M using a digital interpolator. During upsampling, spectral images of the original Nyquist-rate input, which are centered at mf_N for $m = 1, \dots, M-1$, are attenuated with a digital low-pass filter. The resulting oversampled 16-b data is then condensed into a 1-b code by means of a digital $\Sigma\Delta$ modulation noise shaper. Although condensing the 16-b data into 1-b data generates a large amount of quantization noise, the noise shaping algorithm ensures that most of this noise lies above the audio baseband. The oversampled 1-b code at the output of the noise shaper has the same information content within the baseband as the original 16-b input at the Nyquist sampling rate. This 1-b digital signal is converted into an analog signal, and a low-pass filter is then used to remove the out-of-band quantization noise and the remaining spectral images so as to reconstruct the desired baseband analog signal.

As illustrated in Fig. 1, the D/A interface in a conventional oversampling converter is at the output of the noise shaper, where the 1-b digital code is converted directly into a two-level continuous-time signal. A two-level D/A interface is inherently linear because variations in the analog levels only result in a gain error; therefore, the linearity of the interface does not depend on the precise matching of analog circuit components. However, such an interface obviously results in the presence of a large amount of high-frequency quantization noise in the

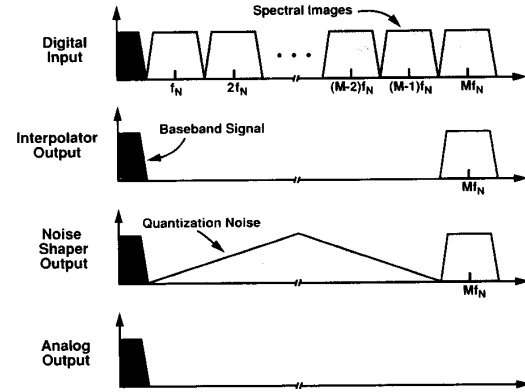


Fig. 3. Spectra of signals at various stages of an oversampling D/A converter.

analog signal, and this noise must be attenuated by the analog reconstruction filter.

If the D/A interface is moved from the output of the noise shaper into the reconstruction filter, as depicted in Fig. 2, additional processing can be performed on the 1-b digital code before it is converted into an analog signal. Instead of generating a two-level analog signal with large quantization noise and then removing the noise in the analog domain, an inherently linear multilevel analog signal is generated with smaller quantization noise. This approach essentially merges the D/A interface with the reconstruction filter to form a semidigital filter.

III. SEMIDIGITAL RECONSTRUCTION FILTER

A. Architecture

The semidigital reconstruction filter in the architecture of Fig. 2 consists of an n -tap semidigital finite-impulse-response (FIR) filter followed by a single-pole low-pass RC filter. A block diagram of the semidigital FIR filter is shown in Fig. 4. The transfer function of this filter is

$$H(z) = \frac{A_{OUT}(z)}{D_{IN}(z)} = a_1 z^{-1} + a_2 z^{-2} + \dots + a_n z^{-n} \quad (1)$$

where the filter coefficients $a_1, a_2, \dots, a_n$ are analog, while the delay elements $z^{-1}, z^{-2}, \dots, z^{-n}$ are implemented digitally using a shift register. The input, D_{IN} , is the 1-b code produced by the digital noise shaper, while the output, A_{OUT} , is a multilevel, continuous-time signal. The transfer function $H(z)$ has a linear phase response if the coefficients are symmetric with

$$a_i = a_{n-i+1} \quad \text{for } i = 1, \dots, n. \quad (2)$$

In the filter of Fig. 4, the digital input D_{IN} is shifted through an n -bit shift register that essentially converts the single-bit serial input into a multibit parallel output. The actual D/A conversion then occurs at the outputs of the shift register. The output at bit i of the shift register, $D_{IN} z^{-i}$, is multiplied by the analog coefficient a_i to generate a weighted, delayed product term. The product terms are analog signals, and these signals are summed to produce a continuous-time output, A_{OUT} . Since

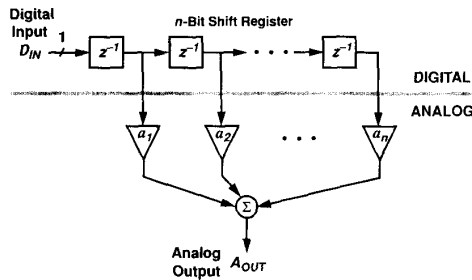


Fig. 4. Semidigital FIR filter.

each product $a_i D_{IN} z^{-i}$ is either a_i or 0, the multiplication can be implemented with a switch controlled by the output of the shift register. The analog output A_{OUT} is simply the sum of all analog coefficients a_i (in voltage, current, or charge) for which the corresponding shift register outputs are at logic ONE.

The semidigital FIR filter can be implemented using either switched-capacitor [6], [7] or switched-current techniques. In a switched-capacitor implementation, the analog coefficients are represented by charges stored on capacitors that can be summed using a switched-capacitor summing amplifier. The individual capacitors in such an implementation need not have low voltage coefficients because the voltage across these capacitors is either 0 or V_{REF} ; thus they can be realized simply with MOS transistors that are appropriately biased. However, the feedback capacitor in the summing amplifier must have a low voltage coefficient in order to ensure that the summing operation meets the linearity requirements of the overall converter.

In this research, a switched-current implementation of the semidigital filter has been adopted because it can be integrated in a conventional digital CMOS technology without the need for on-chip capacitors. In the switched-current realization, which is illustrated in Fig. 5, current sources are used to implement the FIR filter coefficients. The filter thus consists of a shift register, PMOS weighted current sources, CMOS switches, and a current-summing amplifier. The current-summing amplifier provides both current-to-voltage conversion and a single-pole low-pass filtering that "smooths" the output of the overall filter.

B. D/A Interface

The inherent linearity of the semidigital FIR filter can be seen by examining the filter transfer function given in (1). The analog output A_{OUT} is the product of the digital input D_{IN} and the transfer function $H(z)$. The filter is linear if its transfer function $H(z)$, which is implemented with mixed digital and analog circuits, is independent of the digital input D_{IN} .

The digital implementations of the delay and multiplication operations are linear. Unlike delay lines in an analog FIR filter [8], the digital shift register introduces no error in the delay operation itself. Furthermore, performing the multiplications implicitly using CMOS switches avoids input signal dependency in the resulting product terms.

To ensure the converter's linearity, the analog coefficients and the summation of the product terms must also be signal

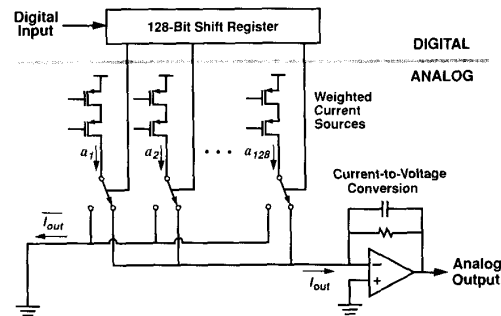


Fig. 5. Current-mode implementation of a semidigital reconstruction filter.

independent. Each of the weighted current sources in Fig. 5 represents an analog coefficient. The current source and its CMOS switch together form a 1-b D/A interface that generates an output current of a_i when the switch is on. The linearity of this D/A interface does not depend on the neighboring current sources. Therefore, mismatch among the analog current sources does not impair either the differential or integral nonlinearity of the converter. Errors in the analog coefficients (current weights), which are similar to coefficient quantization errors in a digital FIR filter [9], do not introduce nonlinearity into the analog output; they simply degrade the stopband response and phase linearity of the filter. However, the current source output a_i must not vary over time or with the signal. The dependency of the current a_i on the output voltage due to channel-length modulation is minimized by summing the currents at the virtual-ground input of a summing amplifier with low harmonic distortion. It is necessary that the summing amplifier itself meet the linearity requirements of the overall converter.

Although the design in Fig. 5 resembles that of a multibit switched-current D/A converter, a multibit converter requires precise matching of the current sources to achieve linearity. The principal difference between the semidigital filter and a multibit D/A converter is that all of the current switches in the semidigital design are controlled by delayed version of the same input signal, whereas in a multibit switched-current D/A converter the control signals are different bit positions of a binary code and thus comprise harmonics of a sinusoidal input signal. Errors in the current weights of a multibit D/A converter will therefore introduce harmonic distortion into the analog output.

C. Discrete-Time to Continuous-Time Interface

In addition to converting digital samples into analog levels, the semidigital reconstruction filter converts discrete-time signals into continuous time. This conversion is accomplished by means of a low-pass filter with a stopband transfer function that adequately attenuates residual spectral images above half the sampling frequency. The design in Fig. 5 combines a zero-order hold function with the single-pole low-pass filter to implement the discrete-to-continuous time conversion. The switched current sources controlled by the shift register act as

the zero-order hold, while the low-pass filter is provided by the RC feedback network in the summing amplifier.

If the zero-order hold and the current-summing low-pass filter are linear and time invariant, then the transfer function $H_{\text{DTCT}}(f)$ for the discrete-to-continuous time conversion consists of a sinc function multiplied by a single pole,

$$H_{\text{DTCT}}(f) = \left[\frac{\sin\left(\frac{\pi f}{f_S}\right)}{\frac{\pi f}{f_S}} \right] e^{-j\frac{\pi f}{f_S}} \left(\frac{1}{1 + j\frac{f}{f_P}} \right) \quad (3)$$

where $f_S = Mf_N$ is the sampling frequency and f_P is the frequency of the low-pass filter pole. The linearity of the conversion does not depend on f_S and f_P as long as these frequencies are time invariant and independent of the signal. However, the relative values of f_S and f_P do influence the frequency response of the filter.

The pole at frequency f_P in the current-summing amplifier of Fig. 5 is chosen so as to adequately suppress residual spectral images without significantly disturbing the baseband frequency response of the converter. From (3) it is apparent that the passband droop introduced by the pole is negligible as long as

$$f_P \gg \frac{f_S}{M} = f_N. \quad (4)$$

For a converter with a large oversampling ratio, this condition can be satisfied even when f_P is less than f_S . For example, if a low-pass filter pole at $f_P = 200$ kHz is used, the maximum passband droop in digital-audio applications is 0.043 dB. Moreover, the passband roll-off due to the zero-order hold is negligibly small when an oversampling ratio of 176 ($f_S = 7.76$ MHz) is used.

In an oversampling D/A converter based on the architecture of Fig. 1, slewing limitations in the generation of the two-level analog output of the D/A interface or in the analog reconstruction filter are potentially troublesome sources of nonlinearity. However, a converter architecture based on the use of semidigital reconstruction filtering is relatively immune to slewing limitations. In the semidigital filter topology of Fig. 5, slewing behavior is a concern only in the operational amplifier used to implement the current-to-voltage conversion. To avoid slewing, this amplifier must have a slew rate larger than the maximum rate of change in the output voltage.

The output voltage response of the reconstruction filter in Fig. 5 is governed by the output current of the semidigital filter and the single-pole RC low-pass filter. The output current of the semidigital filter can change only by a small amount in each clock period because the number of shift register outputs at logic ONE can change only by one per clock period. The largest change in the output current occurs if all of the current source switches are initially in the off position and the digital input is at logic ONE for 128 consecutive clock periods; in that case, one additional current source is switched on per clock period. For such an input, the output current I has a staircase waveform that increases from zero to the full-scale value in 128 clock periods. That is, if switching transients are ignored,

the output current during the k th clock period is

$$I_k(t) = \sum_{j=1}^k a_j, \quad kT \leq t < (k+1)T \quad (5)$$

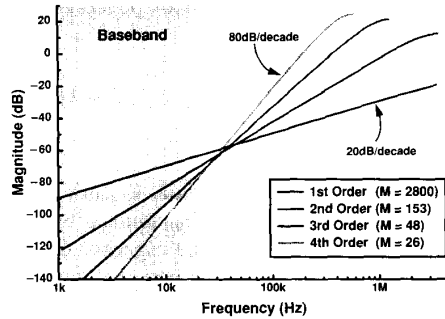
where $T = 1/f_S$ is the clock period. Because the current-to-voltage conversion amplifier is bandlimited by a feedback network consisting of resistor R and capacitor C , the analog output voltage during the k th clock period is given by

$$V_k(t) = V_{k-1}(kT) + \Delta V_k \left(1 - e^{-\frac{t-kT}{RC}} \right), \quad kT \leq t < (k+1)T \quad (6)$$

where $\Delta V_k = [I_k(kT)R - V_{k-1}(kT)]$. From (6) it follows that the maximum rate of change in the output voltage during the k th clock period, which is given by the derivative of $V_k(t)$ with respect to time at $t = kT$, is $\Delta V_k/RC$. The value of ΔV_k can be computed numerically for a given set of filter coefficients using (5) and (6). If the bandwidth of the amplifier's output signal is limited by the single-pole RC feedback network with f_P of 200 kHz, and if the maximum output voltage step size (the maximum value of ΔV_k) is 0.2 V, then to avoid slewing distortion the operational amplifier need only have an output slew rate of at least $2\pi f_P \Delta V_k|_{\text{MAX}}$ or 0.25 V/ μ s.

The use of a return-to-zero coding in a discrete-time to continuous-time interface can ensure that the energy within each clock period is independent of the previous state. This coding scheme is particularly attractive for use in a single-bit discrete-to-continuous time interface with a full-scale output [4] because any nonideality in the rise and fall transitions will only result in a gain error in the output if the shapes of all logic ONE pulses are identical. That is, slewing and unequal rise and fall transition times in the continuous-time output of such a single-bit D/A interface do not affect the linearity of the converter. However, return-to-zero coding is not used in the semidigital reconstruction filter in Fig. 5 because of its relaxed slewing requirements.

Timing jitter in the sampling clock is another source of nonlinearity in the discrete-to-continuous time interface because timing jitter causes f_S in (3) to be time variant. The error due to timing jitter is given by the difference between an ideal continuous-time signal and one with timing jitter. This error is proportional to the amount by which the output changes between adjacent clock periods. As described previously, a semidigital reconstruction filter generates a multilevel analog output that only changes by a small amount between adjacent clock periods. The average error power due to timing jitter in the output of such a filter is therefore much smaller than that in the output of a single-bit discrete-to-continuous time interface, wherein full-scale changes occur between clock periods. Simulations based on the assumption of an uncorrelated random Gaussian error for timing jitter indicate that the worst-case increase in the total inband noise for a -3 dB sinusoidal input at 20 kHz is less than 0.5 dB if the timing jitter has an RMS value less than 500 ps.

Fig. 6. Idealized $\Sigma\Delta$ modulation noise shaping functions.

D. Filter Design

At the output of the noise shaper in an oversampling D/A converter, spectral images at low-order multiples of the Nyquist sampling frequency have already been sufficiently attenuated by the digital interpolation filter. The function of the semidigital reconstruction filter is to attenuate out-of-band quantization noise introduced by the noise shaper and the spectral images that remain at multiples of the oversampling frequency.

The transition-band and stopband requirements for the reconstruction filter are considerably less stringent than those of a digital decimation filter in an oversampling A/D converter because no antialias filtering is needed. In a Nyquist-rate D/A converter, 40–60-dB attenuation of spectral images is considered to be adequate for digital audio applications [10]. In an oversampling D/A converter with a sampling rate of 7.76 MHz, according to (3), the high-frequency spectral images centered at multiples of the sampling frequency are attenuated by at least 80 dB by the combination of the zero-order hold function and a 200-kHz single-pole low-pass filter.

In addition to suppressing the residual spectral images, the reconstruction filter must attenuate the out-of-band quantization noise from the second-order noise shaper. Simulations indicate that for a second-order noise shaper, the total out-of-band quantization noise energy between the baseband edge and $f_s/2$ is approximately 2.6 dB greater than the energy in a full-scale signal. If the total quantization noise energy is attenuated by 63 dB, the out-of-band noise will be at least 60 dB below full scale. In this work, the reconstruction filter attenuates the out-of-band quantization noise energy by 72 dB.

Because the shape of the out-of-band quantization noise depends on the order of noise shaping function, the noise shaper influences the design of the reconstruction filter. Shown in Fig. 6 are plots of the idealized noise shaping functions for a noise differencing $\Sigma\Delta$ modulator, as given by

$$H_L(z) = (1 - z^{-1})^L \quad (7)$$

for $L = 1, \dots, 4$. The oversampling ratio for each noise shaper has been adjusted to give approximately 16-b resolution within the baseband. Although the exact shape of the noise shaping curve may deviate from (7), the curves in Fig. 6 can be used to examine the relationship between the order of the noise shaping and the reconstruction filter.

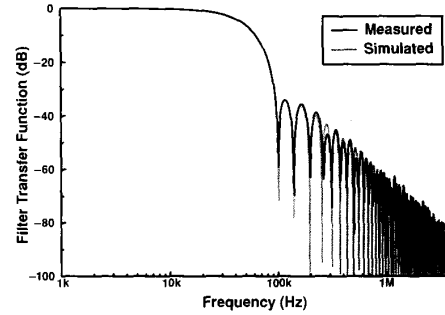


Fig. 7. Measured and simulated frequency response of the reconstruction filter.

It is apparent from Fig. 6 that the slope of the noise shaping function is steeper when the order of noise shaping is higher. If the noise is shaped by a noise differencing of order L , the out-of-band noise increases by approximately $20L$ dB/decade. To attenuate the out-of-band noise, a noise shaper of order L therefore requires the use of a reconstruction filter with an out-of-band attenuation that falls by at least $-20L$ dB/decade.

If an analog reconstruction filter is used to attenuate the out-of-band noise generated by a noise shaper with single-bit quantization, its order must be equal to or greater than the order of noise shaping. For example, a second-order noise shaper requires the use of a reconstruction filter with an out-of-band transfer characteristic that falls by at least -40 dB/decade; that is, it must at least be a second-order low-pass filter. In order to simplify the analog reconstruction filter, the lowest possible order of noise shaping function is therefore preferred.

The tradeoff between the order of noise shaping and the complexity of an analog reconstruction filter is avoided if a semidigital reconstruction filter is used. An FIR filter requires more taps to obtain a larger stopband attenuation but fewer taps if the transition bandwidth, normalized to the oversampling frequency, is wider. For high-order noise shaping with a low oversampling ratio, the increase in the number of taps due to the need for greater stopband attenuation is offset by a reduction due to the wider normalized transition bandwidth. Thus, a high-order noise shaper with a low oversampling ratio does not necessarily require a larger semidigital reconstruction filter.

The design of digital FIR filters has been the subject of extensive research [9]. Standard digital filter design techniques are directly applicable to semidigital FIR filters. In this work, the analog filter coefficients have been established using the frequency sampling method [9]. The number of taps in the FIR filter were minimized to the extent possible in order to conserve die area. Fig. 7 shows the transfer function of a semidigital reconstruction filter that has been designed for use with a second-order noise shaper. This filter consists of a 128-tap semidigital FIR filter followed by a 200-kHz single-pole low-pass filter. The stopband envelope of the FIR filter has a slope of -20 dB/decade, and its attenuation at $f_s/2$ is 55 dB. The pass-band droop is 0.89 dB. The composite semidigital reconstruction filter (combining both the FIR and 200 kHz low-pass filters) has an out-of band attenuation slope of -40

dB/decade and attenuates the out-of-band quantization noise energy between 20 kHz and $f_S/2$ (3.88 MHz) by 72 dB.

E. Filter Coefficient Error

A digital implementation of an FIR filter has deterministic quantization errors, whereas in a semidigital filter the analog coefficients are subject to less predictable errors. The following analysis describes how random errors in the analog coefficients can affect the transfer function of a reconstruction filter.

If it is assumed that the coefficient errors in an n -tap FIR filter, with $H(f) = 1$ within the passband, are uncorrelated Gaussian random variables with a mean of zero and a standard deviation σ_E , then the error in the magnitude of the transfer function $H(f)$ can be approximated by [11]

$$|\Delta H(f)| \approx \frac{\sqrt{\pi n}}{2} \sigma_E. \quad (8)$$

For the switched-current implementation shown in Fig. 5, all of the current sources are biased identically and have the same gate length but different gate widths. An estimate of σ_E can be obtained by assuming that the current source errors are random with an average standard deviation δ_I . The number of taps used in the semidigital filter does not affect δ_I provided that the average value of the current sources is kept at I_M . For an n -tap filter with a full-scale output current of nI_M , the standard deviation of the output current error is $\delta_I \sqrt{n}$. Therefore, the standard deviation of coefficient errors for an output normalized to give $H(f) = 1$ within the passband, when expressed as a fraction of the full-scale signal, is

$$\sigma_E \approx \frac{\delta_I}{I_M \sqrt{n}}. \quad (9)$$

Upon substitution of (9) into (8) it follows that

$$|\Delta H(f)| \approx \frac{\sqrt{\pi}}{2} \left(\frac{\delta_I}{I_M} \right). \quad (10)$$

(10) indicates that the transfer function error $|\Delta H(f)|$ in a current-mode semidigital FIR filter is, to first order, independent of frequency and the number of taps in the filter. Since δ_I/I_M is typically much less than unity, random errors in the current sources have a negligible effect within the passband of the filter. The current source errors can, however, limit the maximum attenuation of the normalized stopband envelope to approximately $1/|\Delta H(f)|$.

The effect of current source errors on the out-of-band attenuation can be estimated from (10). If the average current source has a maximum error of $2^{-7} I_M$ (6-bit accuracy), and it is assumed that the standard deviation δ_I is one-fifth of the maximum error, then the passband and stopband errors are approximately 0.01 dB and -57 dB, respectively. The 0.01-dB passband error is indeed negligible in most applications.

The stopband response of the filter with the transfer function of Fig. 7 is relatively insensitive to random coefficient errors because the FIR filter has a maximum attenuation of only 55 dB in its stopband envelope. The additional -20-dB/decade roll-off from the 200-kHz pole in the current-summing amplifier provides for the attenuation of the total out-of-band

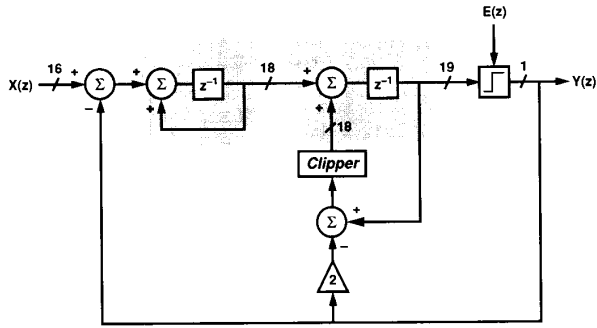


Fig. 8. Z-domain block diagram of a second-order noise shaper.

quantization noise energy by 72 dB. Computer simulations indicate that this filter design will attenuate the out-of-band noise by at least 70 dB if random errors in the coefficients are no larger than 2^{-5} times the average value of the coefficients.

IV. IMPLEMENTATION

A. Interpolator

The purpose of the digital interpolator in oversampling D/A converters of the form shown in Figs. 1 and 2 is to increase the sampling frequency of the Nyquist-rate digital input signal and attenuate the spectral images associated with the upsampling process. This interpolation is usually implemented in two or more steps [1]. For digital-audio applications the Nyquist-rate signal is typically first upsampled by two to eight times using a digital interpolation FIR filter with a sharp transition band from 20 kHz to 24 kHz and a stopband attenuation of 60–80 dB. Additional increases in sampling rate can then be accomplished using linear interpolation since the remaining spectral images are sufficiently attenuated by the interpolation to be buried in the shaped quantization noise produced by the noise shaper.

The experimental prototypes implemented in this work included a linear interpolator but not the digital interpolation FIR filter. Instead, a ROM-based digital sinewave generator [12] was used to provide an input at 11 times oversampling to the device under test. The on-chip linear interpolator, which is similar to the design described in [13], increases the sampling rate by an additional factor of 16 to generate the oversampled input to the digital noise shaper.

B. Noise Shaper

The digital noise shaper condenses the 16-b upsampled digital signal from the interpolator into a 1-b code with 16-b baseband resolution. A second-order noise shaper with single-bit quantization and an oversampling ratio of 176 has been used in the experimental implementation primarily because of its simplicity. A z -domain block diagram of the second-order noise shaper, which is similar to that for a conventional $\Sigma\Delta$ modulator used for A/D conversion, is shown in Fig. 8 [14]. It consists of two digital integrators, a quantizer, a gain of two, a clipper, and two subtractors. Because two's complement arithmetic is used, the quantizer and gain block

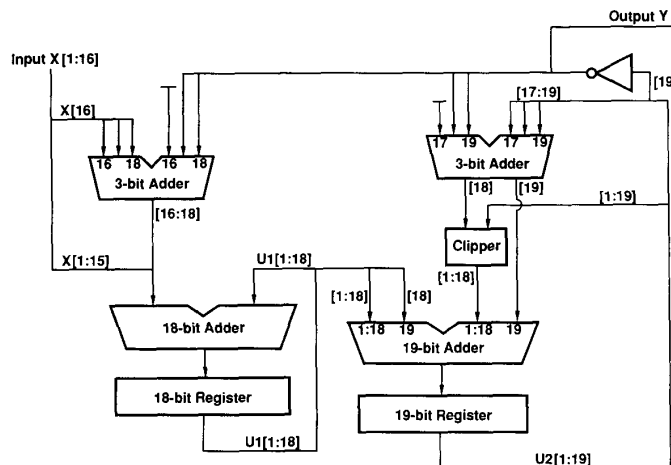


Fig. 9. Circuit implementation of a second-order noise shaper.

need not be implemented explicitly. In addition, if the positive and negative full-scale levels at the output of the quantizer are chosen to be 08000_{HEX} and 18000_{HEX} so that in both cases the lower 15 bits are zero, then only the three most significant bits in the two subtractors need be implemented.

Fig. 9 shows the circuit implementation of the second-order noise shaper. The quantizer output is simply the sign-bit from the second-stage output. Two 3-b adders, each with one input complemented, implement the two subtraction operations. The gain of two is wired as a left shift by one bit at the input of the second subtractor. The clipper is implemented using combinational logic. With ripple-carry adders and with registers implemented using master-slave latches, the noise shaper will operate at clock rates as high as 20 MHz, which is more than twice the required sampling frequency of 7.76 MHz.

Computer simulations have been used to determine the data wordlength needed within the each stage of the noise shaper to provide a theoretical dynamic range of at least 98 dB for sinusoidal signals. The wordlengths in the first and second stages are 18 b and 19 b, respectively. These lengths allow the noise shaper to operate without roundoff error for input signals as large as 2 dB below full scale. However, when the input signal level exceeds -2 dB, the signal range within the noise shaper increases dramatically. A clipper is therefore used in the second stage to avoid wrap-around of the two's complement numbers.

Quantization error at the output of the noise shaper is not random; it may include low-amplitude harmonics, as well as spurious noise tones. Because the uniform quantization of a sinewave results in noise that is purely discrete and consists of odd harmonics of the fundamental [15], a Nyquist-rate digital sinusoidal input obtained from such quantization contains low-amplitude harmonics that may appear in the output of the noise shaper, especially if the sampling frequency is an integer multiple of the input signal frequency. Noise tones and low-level harmonics may also be generated by the second-order $\Sigma\Delta$ noise shaper for dc and low-amplitude inputs as a result of limit-cycle oscillations [16]. The harmonics and noise tones

are typically below the quantization noise corresponding to 16-b resolution and are potentially observable only for low-amplitude and dc inputs. These tones can be further suppressed by adding a dither signal to the input or by using a higher order noise shaper [15], [17].

C. Reconstruction Filter

The current-mode reconstruction filter shown in Fig. 5 consists of a 128-b shift register that delays the digital input D_{IN} , 128 weighted PMOS current sources that represent analog FIR filter coefficients, 128 pairs of CMOS switches that implement the multiplication functions, and an external summing amplifier with a pole at 200 kHz that provides current-to-voltage conversion and filters high-frequency spectral images.

The principal focus of the circuit design for the reconstruction filter was that of minimizing signal dependency in the weighted current sources and the current summing operation. Relatively large gate lengths of $18 \mu\text{m}$ are used for the PMOS current source transistors to minimize flicker and thermal noise, as well as reduce channel-length modulation. The widths of the PMOS current-source transistors, which vary from $13.8 \mu\text{m}$ to $124.8 \mu\text{m}$, are proportional to the FIR filter coefficients, a_1 through a_n , and are quantized in increments of $0.6 \mu\text{m}$. A large gate-to-source bias voltage of 2 V is used to minimize the sensitivity of the weighted currents to transistor threshold variations and fluctuations in the gate voltage due to switching noise. The current sources are biased to provide a full-scale output current of ± 1 mA. Cascode transistors are used to increase the output resistances of the current sources and buffer the voltage reference line that biases the current sources from transients in the current switches. The individual current sources are switched between the virtual ground and ground through CMOS switches that operate in the linear region. These switches are driven by crossed-coupled NAND gates with overlapping on-times to ensure that the current sources always have a path to ground. The filter output current,

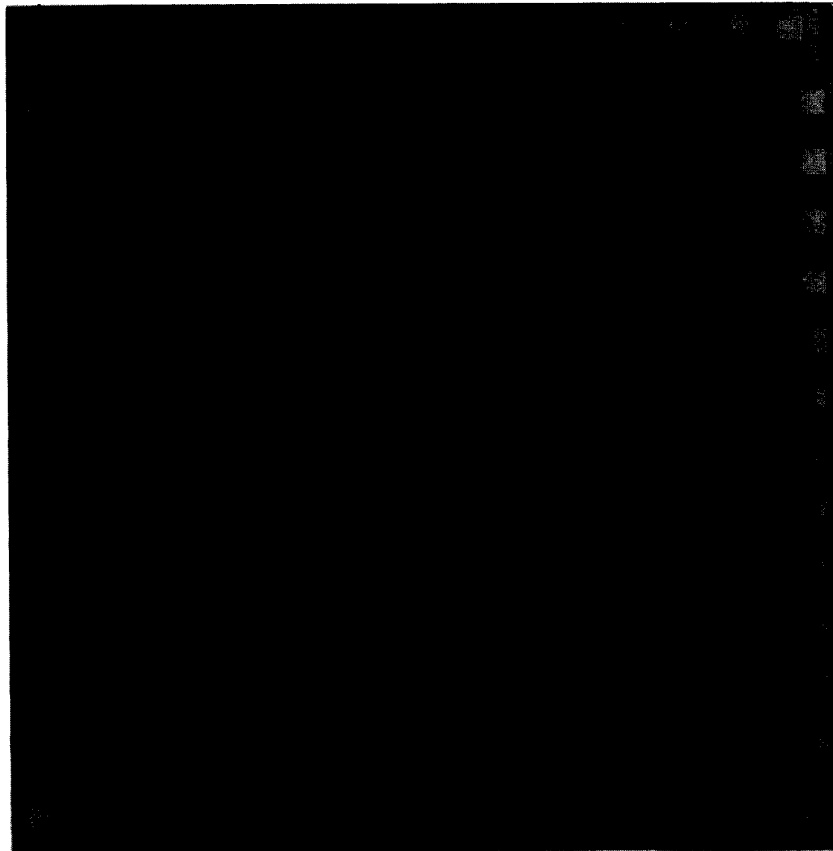


Fig. 10. Die photo of experimental prototype.

I_{OUT} , is summed at the virtual-ground input of the inverting operational amplifier.

D. Layout

An experimental prototype consisting of a linear interpolator, a second-order noise shaper, and a 128-tap semidigital FIR filter has been fabricated in a 1.2- μm CMOS technology. A die photo of this circuit is shown in Fig. 10. The linear interpolator, which is in the lower right corner of the figure, has an active area of 0.38 mm². The second-order noise shaper is located in the upper right corner of the die and has an active area of 0.64 mm². The semidigital FIR filter occupies the left half of the die and has an active area of 1.98 mm². The 128 current sources are laid out as two rows of 64 current sources, corresponding to the two symmetric halves of the impulse response, that are mirrored along a vertical axis and surrounded by the current switches and shift register.

The test chips were packaged in 44-pin J-leaded chip carriers. Special routing and packaging steps were taken to minimize the coupling of digital switching noise into sensitive analog circuits via the supplies and substrate. Separate bond pads are used for the analog and digital supplies. Instead of biasing the substrate using signal grounds, all substrate contacts are routed separately to dedicated bond pads. Multiple

package pins are used for all supply and substrate connections to reduce the parasitic inductance of the bond wires and package traces.

V. EXPERIMENTAL RESULTS

A. Test Setup

Special care was taken in the test setup to minimize the coupling of digital switching noise into the analog output. A two-sided copper board was used with all interconnects routed on the top side and separate analog and digital ground planes on the bottom side. The digital and analog grounds are connected to their respective ground planes. Within the analog section of the board, all current-carrying ground returns are routed separately and connected together at a common ground point. A low-lead inductance surface-mount PLCC socket was used. All supply pins are bypassed with 0.01- μF chip capacitors at the feet of the socket. The substrate pins are connected to the analog ground plane. All digital inputs to the device under test are optically coupled to eliminate potential ground loops and the mixing of digital ground currents into the analog signal paths.

A digital sinewave generator provided an upsampled 16-b input at 485.1 kS/s (11 times oversampling) to the device

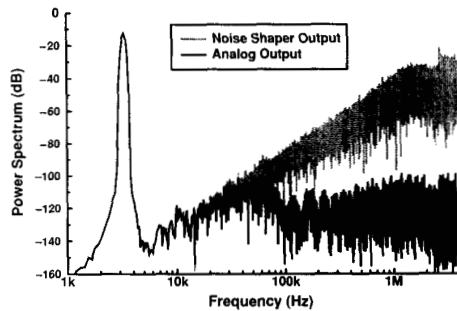


Fig. 11. Power spectrum of a 3.31-kHz signal before and after reconstruction filtering.

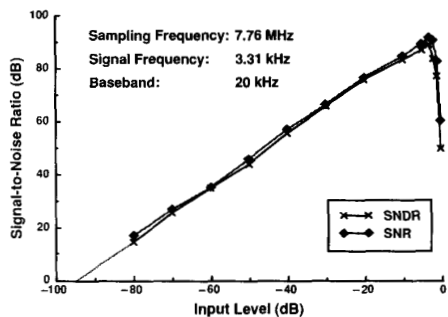


Fig. 12. Measured SNR and SNDR.

under test. Conversion of the output current to a voltage and the single-pole low-pass filtering were implemented using an external operational amplifier (OP27) with a feedback network consisting of a 1-k Ω metal-film resistor and a 800-pF NPO-type capacitor. The analog output voltage was analyzed using a digital spectrum analyzer with a fast Fourier transform (FFT) capability.

B. Measured Results

Fig. 7 shows the measured and simulated transfer functions of the reconstruction filter. The transfer function was obtained by taking the Fourier transform of an impulse response. Close agreement between the measured and simulated responses was achieved. The measured deviation from linear phase of 0.03 $^\circ$ is due almost entirely to phase error from the 200-kHz low-pass filter. Since the measured frequency response agrees well with the simulated response within the passband, the passband droop of 0.93 dB can be corrected digitally in the interpolator.

The power spectrum of a sinusoidal signal before and after filtering has been simulated based on a fit of the measured filter transfer function in Fig. 7. A 65 536-point FFT-based power spectrum, with Nuttall windowing [18], of the output response to a 3.31-kHz sinusoidal input 10-dB below overload is shown in Fig. 11. At the output of the noise shaper, the out-of-band noise increases at about 40 dB/decade since second-order noise shaping is used. The semidigital reconstruction filter reduces the total out-of-band power between 20 kHz and 3.88 MHz ($f_s/2$) by 72 dB.

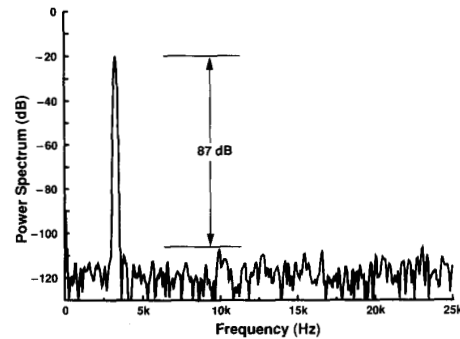


Fig. 13. Measured power spectrum of a -20-dB signal.

TABLE I
PERFORMANCE SUMMARY

Noise Shaping	Second order
Oversampling Ratio	176
Dynamic Range	94 dB
Peak SNDR	88 dB
Power Dissipation (5 V and ± 1 mA output current)	59 mW
Out-of-Band Attenuation	72 dB
Active Die Area (1.2- μ m CMOS)	3 mm ²

To measure the linearity and dynamic range of the converter, the analog output was analyzed at various amplitudes using the FFT-based spectrum analyzer. The measured signal-to-noise ratio (SNR) and signal-to-noise+distortion ratio (SNDR) of the experimental converter are plotted as functions of the input level in Fig. 12. This figure indicates that the dynamic range of the converter is 94 dB, with a peak SNR of 91 dB and a peak SNDR of 88 dB. The measured SNDR of the converter is about 4–6 dB less than the theoretical performance of the digital noise shaper. The harmonic distortion and noise floor of the measured power spectra were sensitive to board layout, grounding configurations, and the digital supply voltage. The primary source of the distortion appears to be the coupling of signal-dependent digital switching noise into the analog current source reference. This switching noise was minimized by operating the digital circuits from a 3.3-V supply.

Fig. 13 shows the measured output power spectrum of a -20-dB sinewave at 3.31 kHz based on a 1024-point FFT (400 frequency bins) with Blackman-Harris windowing [19]. The harmonics and noise floor of the measured spectrum are 87 dB below the fundamental. For a baseband of 20 kHz, the -20-dB signal has a SNDR of 75 dB. The performance of the experimental converter is summarized in Table I.

VI. CONCLUSION

An audio-band oversampling D/A converter with semidigital reconstruction filtering has been designed and integrated in a 1.2- μ m CMOS technology. With a second-order noise shaper and an oversampling ratio of 176, the D/A converter has a dynamic range of 94 dB, a peak SNR of 91 dB, and a peak SNDR of 88 dB. The out-of-band quantization noise power between 20 kHz and 3.88 MHz is attenuated by 72 dB. The power dissipation of the experimental circuit, when

operating from a 5-V supply, is 59 mW, and the total active die area is 3 mm².

The experimental results show that semidigital reconstruction filtering is a feasible alternative to more conventional switched-capacitor or RC filters for oversampling D/A conversion. A semidigital filter combined with single-bit noise shaping provides a multilevel D/A interface with low quantization noise, without requiring stringent component matching. The use of current-switching circuits allows the filter to be implemented in a standard digital CMOS technology without the need for high-quality on-chip capacitors. In addition, it may be possible to operate such a current-mode D/A converter at reduced supply voltages for low-power applications.

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